



PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR UNIVERSITY, SOLAPUR



IMPORTANT INSTRUCTION: READ ALL THE INSTRUCTIONS CAREFULLY BEFORE FILLING THE INFORMATION
 • This Sheet will be scanned by Computer • USE DARK BLUE PEN for writing the Seat Number and other Codes.
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 in BLUE INK PEN

READ THE INSTRUCTION ON THE BACK SIDE

PART-I

STUDENT TO FILL THIS AREA FIRST

Examination : Oct./Nov. - 2025 Branch: Mechanical Engg Medium: English
 Subject Name & Code : Basics of Electric vehicle (EEMDM-01B)
 Year/Part : S.Y. Semester : 3rd Paper No. : - Section : Z & II
 Q. Paper Code/SLR No. JE - 85 Set No. 9 Date 24/12/2025 Time : 8.00 - 6.00

FOR OFFICE USE ONLY

Q. No.	1	2	3	4	5	6	7	TOTAL
Examiner								
Moderator								
2nd Examiner								
Name & Signature	Examiner		Moderator		2nd Examiner			

Answer Booklet No. Barcode



25493362

ही जागा परीक्षकांनी गुणदान करण्यासाठी आहे. विद्यार्थ्यांनी येथे काहीही लिहू नये

PART-II

Examiner			Moderator			2nd Examiner			Name Code & Signature		Answer Booklet Number
0	0	0	0	0	0	0	0	0	Examiner / परीक्षक	 25493362	
1	1	1	1	1	1	1	1	1	Moderator / नियमक		
2	2	2	2	2	2	2	2	2			
3	3	3	3	3	3	3	3	3			
4	4	4	4	4	4	4	4	4			
5	5	5	5	5	5	5	5	5			
6	6	6	6	6	6	6	6	6			
7	7	7	7	7	7	7	7	7			
8	8	8	8	8	8	8	8	8			
9	9	9	9	9	9	9	9	9			

IE 85 202874

READ THE INSTRUCTION ON THE BACK

PART-III

STUDENT TO FILL THIS AREA FIRST

Examination : Oct./Nov. - 2025 Branch: ME Medium: English
 Seat No. in words Two Lakh Three Thousand Eight Hundred Sixty - seven

FILL YOUR SEAT NUMBER

2	0	3	8	6	7
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Subject Name & Code : Basics of Electric vehicle (EEMDM-01B)

Year / Part : S.Y. Semester : 3rd Paper No. : - Section : Z & II

Exam Centre : BMZT, Solapur

Q. Paper Code / SLR No.

JE - 85

Set No. (P/Q/R/S) - 9

Date : 24/12/2025

Time : 3:00 - 6:00

Answer Booklet Number



25493362

Pohan

Candidate Signature

M.D. Chavan

Junior Supervisor Name & Signature

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PART-I

STUDENT TO FILL THIS AREA FIRST

Examination : Oct./Nov. - 2025 Branch: Mechanical Engg Medium: English
 Subject Name & Code : Basics of Electric vehicle (EEMDM-018)
 Year/Part : S.Y. Semester : 3rd Paper No. : - Section : Z & II
 Q. Paper Code/SLR No. JE - 85 Set No. Q Date 24/12/2025 Time : 8.00 - 6.00

FOR OFFICE USE ONLY								
Q. No.	1	2	3	4	5	6	7	TOTAL
Examiner								
Moderator								
2nd Examiner								
Name & Signature	Examiner		Moderator		2nd Examiner			

Answer Booklet No. Barcode



25493362

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PART-II

Examiner	Moderator	2nd Examiner	Name Code & Signature	Answer Booklet Number
0 0 0	0 0 0	0 0 0	Examiner / परीक्षक	
1 1 1	1 1 1	1 1 1		
2 2 2	2 2 2	2 2 2	Moderator / नियमक	
3 3 3	3 3 3	3 3 3		
4 4 4	4 4 4	4 4 4		
5 5 5	5 5 5	5 5 5		
6 6 6	6 6 6	6 6 6		
7 7 7	7 7 7	7 7 7		
8 8 8	8 8 8	8 8 8		
9 9 9	9 9 9	9 9 9		



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IE 85 202874

READ THE INSTRUCTION ON THE BACK

PART-III

STUDENT TO FILL THIS AREA FIRST.

Examination : Oct./Nov. - 2025 Branch: ME Medium: English
 Seat No. in words Two Lakh Three Thousand Eight Hundred Sixty - seven

FILL YOUR SEAT NUMBER

2	0	3	8	6	7
0 0 0 0 0 0	1 1 1 1 1 1	2 2 2 2 2 2	3 3 3 3 3 3	4 4 4 4 4 4	5 5 5 5 5 5
6 6 6 6 6 6	7 7 7 7 7 7	8 8 8 8 8 8	9 9 9 9 9 9		

Subject Name & Code : Basics of Electric vehicle (EEMDM-018)

Year / Part : S.Y. Semester : 3rd Paper No. : - Section : Z & II

Exam Centre : BMZT, Solapur

Q. Paper Code / SLR No.

JE - 85

Set No. (P/Q/R/S) - Q

Date 24/12/2025

Time : 3:00 - 6:00

Answer Booklet Number



25493362

Pohan

Candidate Signature

M.D. Chavan

Junior Supervisor Name & Signature

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SPECIAL INSTRUCTIONS TO CANDIDATES

- Enter on the cover page of each of the answer - books, your examination/Seat number, the name of the examination the date the subject of the question paper and the language of the answer.
- Do not write your Seat No. at any place in the answer book except in the space provided for it.
- DO NOT REVEAL YOUR IDENTITY IN ANY FORM IN THE ANSWERS WRITTEN BY YOU OR AT ANY OTHER PART OF THE ANSWER BOOK, WRITING YOUR NUMBER OR NAME PUTTING YOUR SIGNATURE IS REVELATION OF IDENTITY. USE OF RELIGIOUS INVOCATION OR ANY WRITING THAT IS NOT RELEVANT TO THE ANSWERS OUTSIDE OR INSIDE OF THE BODY OF THE ANSWERS WILL BE TREATED AS ATTEMPT TO REVEAL IDENTITY.
- Margins are provided on the page. Candidate should not draw additional margins. He/She should not write anything in the margin.
- A) Candidates are forbidden to (i) Bring any books, notes, mobile phones any electronic gadgets or scribbling papers into the examination hall (ii) Speak or communicate in any manner to other candidate while the examination is in progress and (iii) Take with them any answer booklet written or blank while leaving the examination hall.

B) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after they give an undertaking in writing that the decision of the University in respect of the reported act of unfair means is binding on them.

C) Candidates are requested to go through the provisions of Maharashtra Act No. XXXI of 1982 i.e. An Act to provide for preventing malpractice at University Board and other specified examinations, published in the Maharashtra Government gazettes dated 14th Oct. 1982. Every offence under this Act is cognizable and non-bailable.
- Candidates should write their answers in a Legible hand. They are warned that no marks will be assigned for answers which cannot be assessed by the examiners owing to illegible handwriting
- Begin your answer of each question on a new page for each answer write the corresponding question number in the margin. Use the paper judiciously. Do not waste the pages. Write on both side of the pages of the answer book.
- This answer book contains 18 pages excluding the cover page. Therefore the students are requested to answer within these pages only.
- Supplement will be provided.**
- Use only **Blue ink Ball point pen to shade the bubbles.**

11. As per the sample shown below write your Seat No. & fill the circles accordingly using **Blue ink ball pen** for e.g. If the Student's Seat No. is 123456 write in figures in the square boxes and fill the circles as shown in the box.

12. Write the respective Set.No. (P/Q/R/S) that you are solving on Cover Page on Part-1 Main Slip and Part-III Code Slip and on Third Page of Answer book.

13. If Candidate does not writes their respective Set No. i.e P/Q/R/S, then assesment shall be done as per Set No. P.

14. If Candidate writes multiple answer of MCQ then such answer of MCQ will not be considered and student will get zero marks for it.

15. Write the objective type questions on Page No. 3 only.

N.B.: A candidate who disobeys any instruction issued by the Centre Incharge/Centre Observer/Senior/Junior Supervisor or who is guilty or rude or disobedient is liable for disciplinary action to be taken against him/her by the University.

FILL YOUR SEAT NUMBER						
1	2	3	4	5	6	
0	☐	☐	☐	☐	☐	☐
1	☒	☐	☐	☐	☐	☐
2	☐	☒	☐	☐	☐	☐
3	☐	☐	☒	☐	☐	☐
4	☐	☐	☐	☒	☐	☐
5	☐	☐	☐	☐	☒	☐
6	☐	☐	☐	☐	☐	☒
7	☐	☐	☐	☐	☐	☐
8	☐	☐	☐	☐	☐	☐
9	☐	☐	☐	☐	☐	☐



Q.No.

Examiner

Moderator

Answer Objective Questions only here

a	b	c	d	e	=	

Set. No. : P/Q/R/S

3

MCQ/objective type Questions

Q.1

Q.No	Ans	Q.No	Ans
1.	a]	8.	a]
2.	a]	9.	d]
3.	c]	10.	b]
4.	c]	11.	c]
5.	d]	12.	c]
6.	b]	13.	d]
7.	a]	14.	d]

Q.No.
Examiner

Moderator

a	b	c	d	e	=	
					=	
					=	

Set. No. : P/Q/R/S

4

Section - I

Q.2 a]

Answer

Sl. No.	electric vehicles	ICE vehicles
1.	energy source is Battery	energy source is fuel (petrol & Diesel)
2.	Maintenance cost is low low	Maintenance cost is High
3.	Zero pollution	High pollution
4.	³ Q(2.1) low efficiency	High efficiency
5.	No No Noise pollution	Noise pollution

Q.No.
Examiner

Moderator

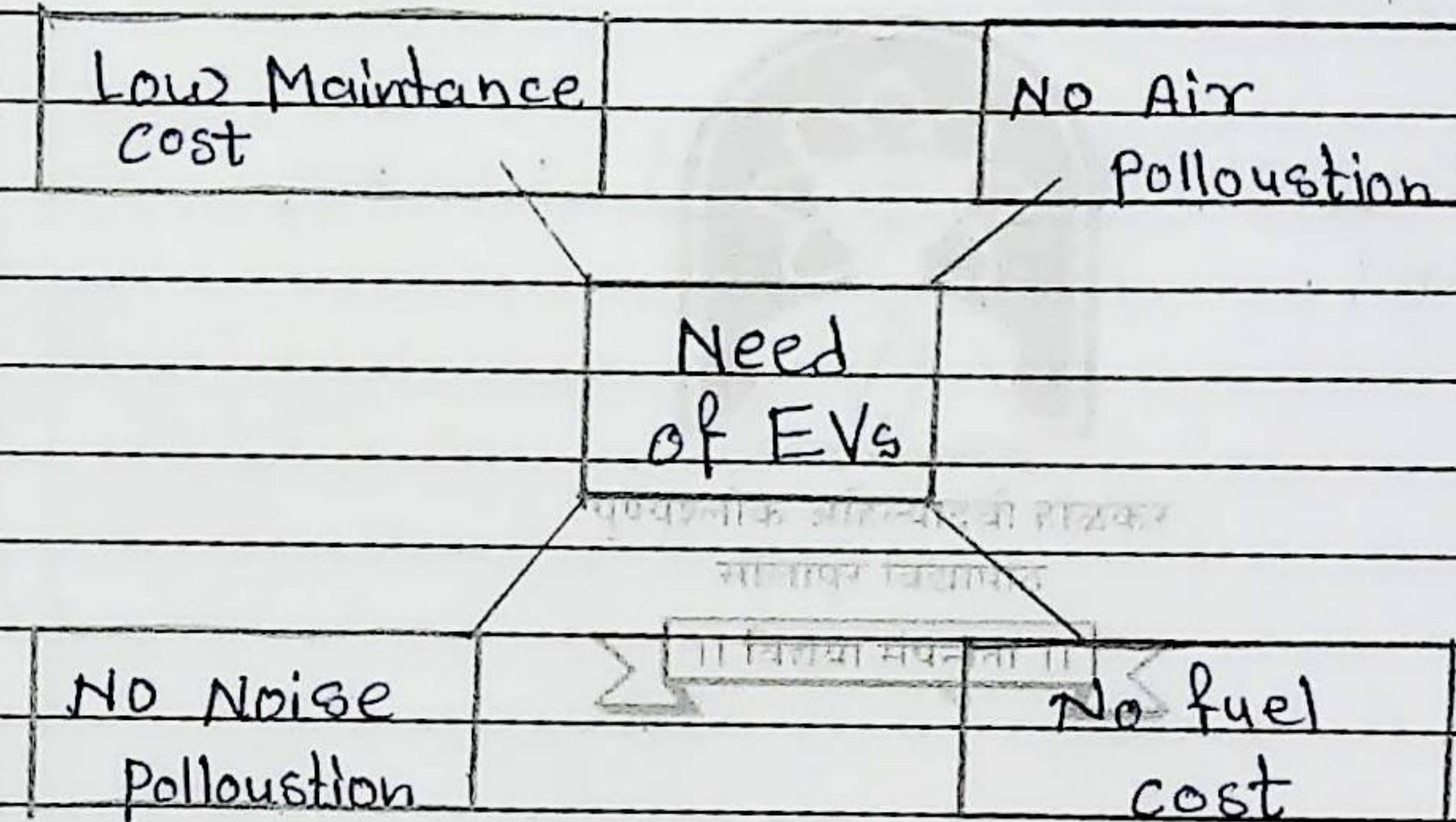
a	b	c	d	e	=	
					=	
					=	

Set. No. : P/Q/R/S

5

Q.2 c]

Answer



a] Low Maintenance →

Electric vehicle is energy source is Battery, not using fuels (petrol & Diesel) and No engine oil and not clutch.

b] No Noise pollution →

Electric vehicle is run by smoothly and NO voices so as NO Noise pollution.

Q.No.

Examiner

a	b	c	d	e	=	

Moderator

Set. No. : P/Q/R/S

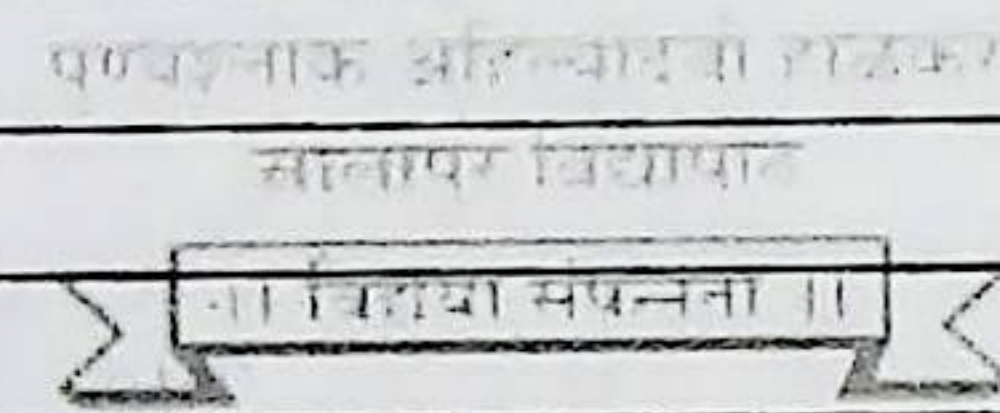
6

c] No Air pollution →

Electric ^{Q(2.3)} vehicle as not use to fuel.
So no exust gas and not Exust system.

d] No fuel cost →

Electric vehicle an not using
only fuel.

Q.No.

Examiner

a	b	c	d	e	=	

Moderator

Set. No. : P/Q/R/S

7

Q.2 d]

Answer

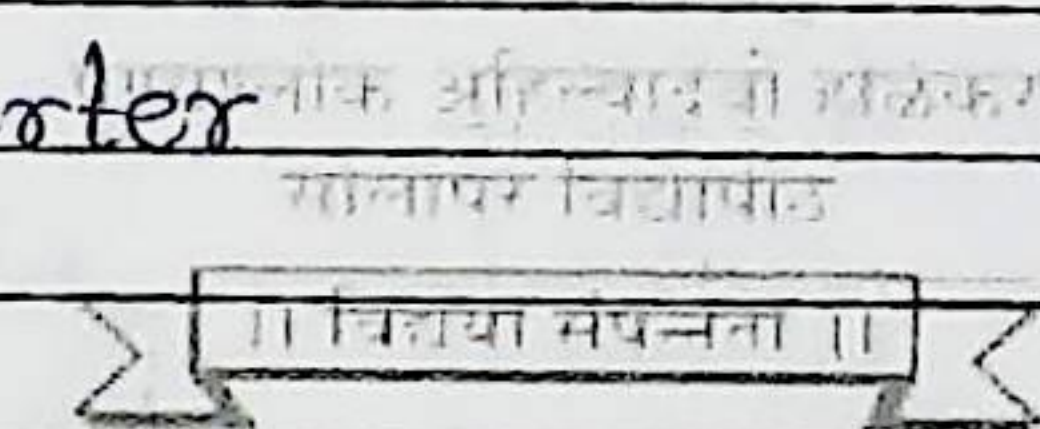
a] Transmission System

b] DC-DC system

c] ^{Q(2.4)} power Generation

d] Braking system

e] power converter



Q.No.

Examiner

a	b	c	d	e	=	

Moderator

Set. No. : P/Q/R/S

8

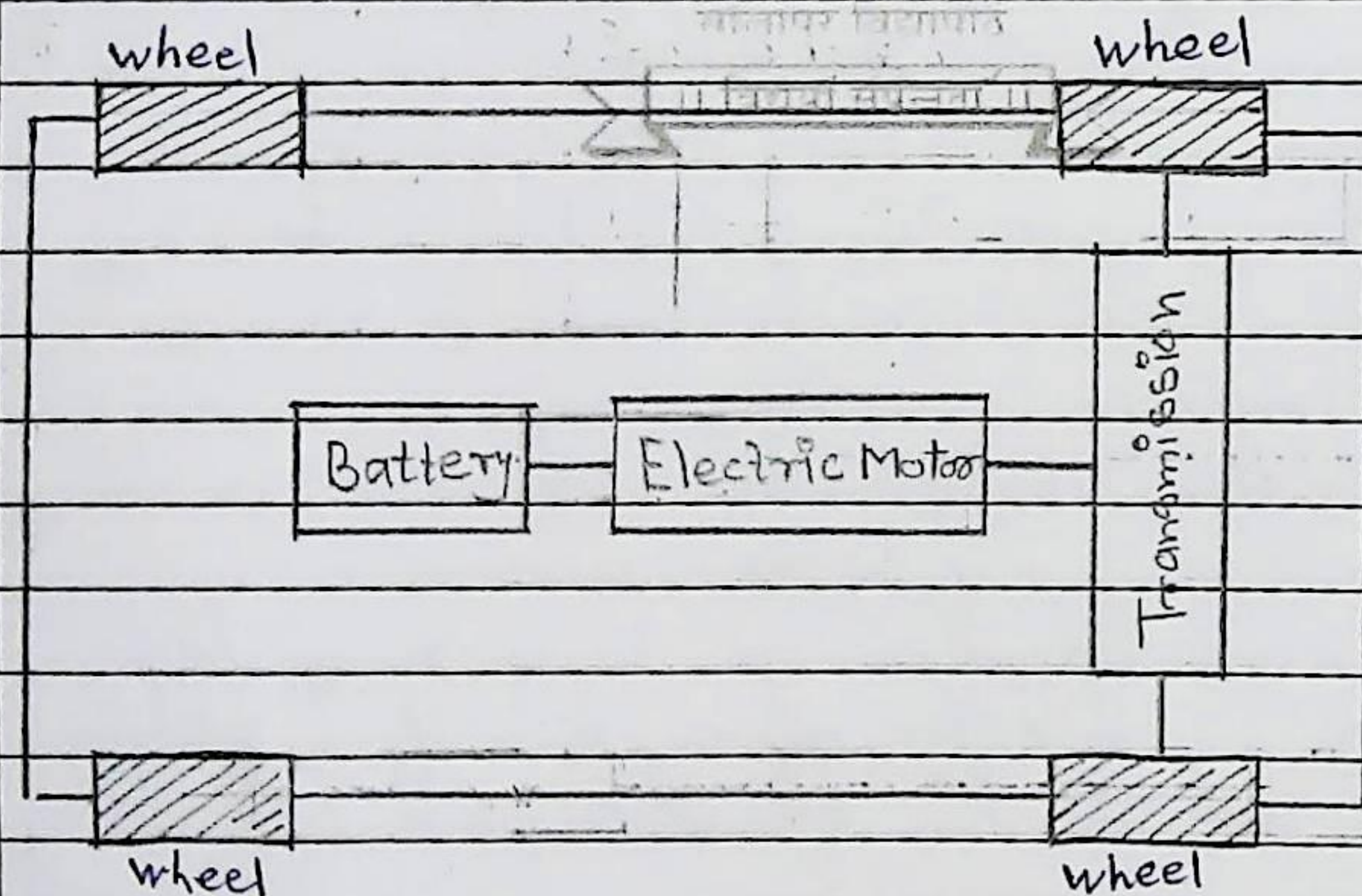
Q.3 a)

Answer

* Types of Electric vehicles :-

- Battery Electric vehicle
- Hybrid Electric vehicle
- plug-in Hybrid Electric vehicle
- Fuel cell Electric vehicle

a) Battery Electric Vehicle



- Battery Electric Vehicle is grid and Home on Board charging vehicle.

- First to charge the Battery then Battery source in pass the Electric motor.

Q.No.

Examiner

a	b	c	d	e	=	

Moderator

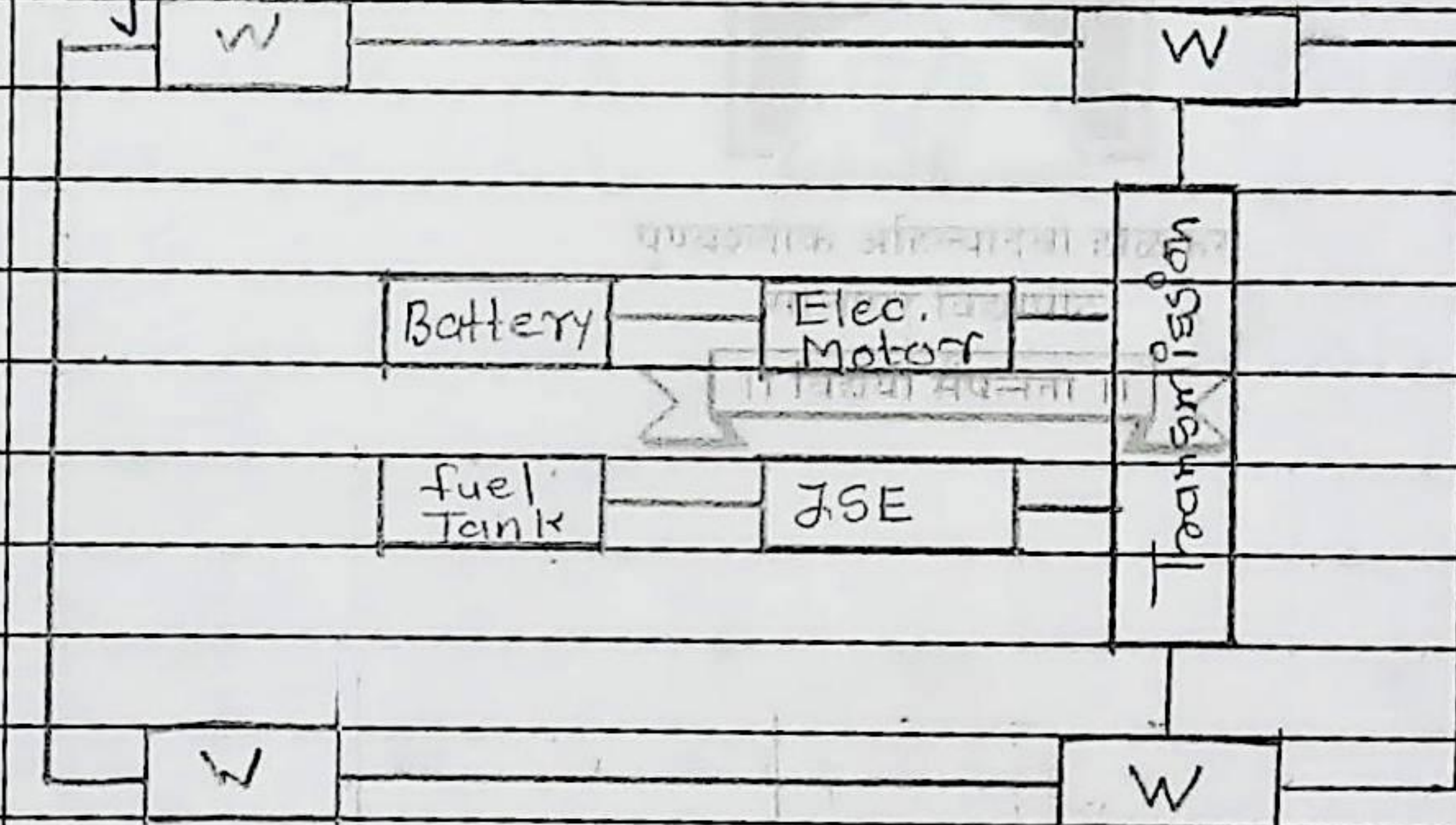
Set. No. : P/Q/R/S

9

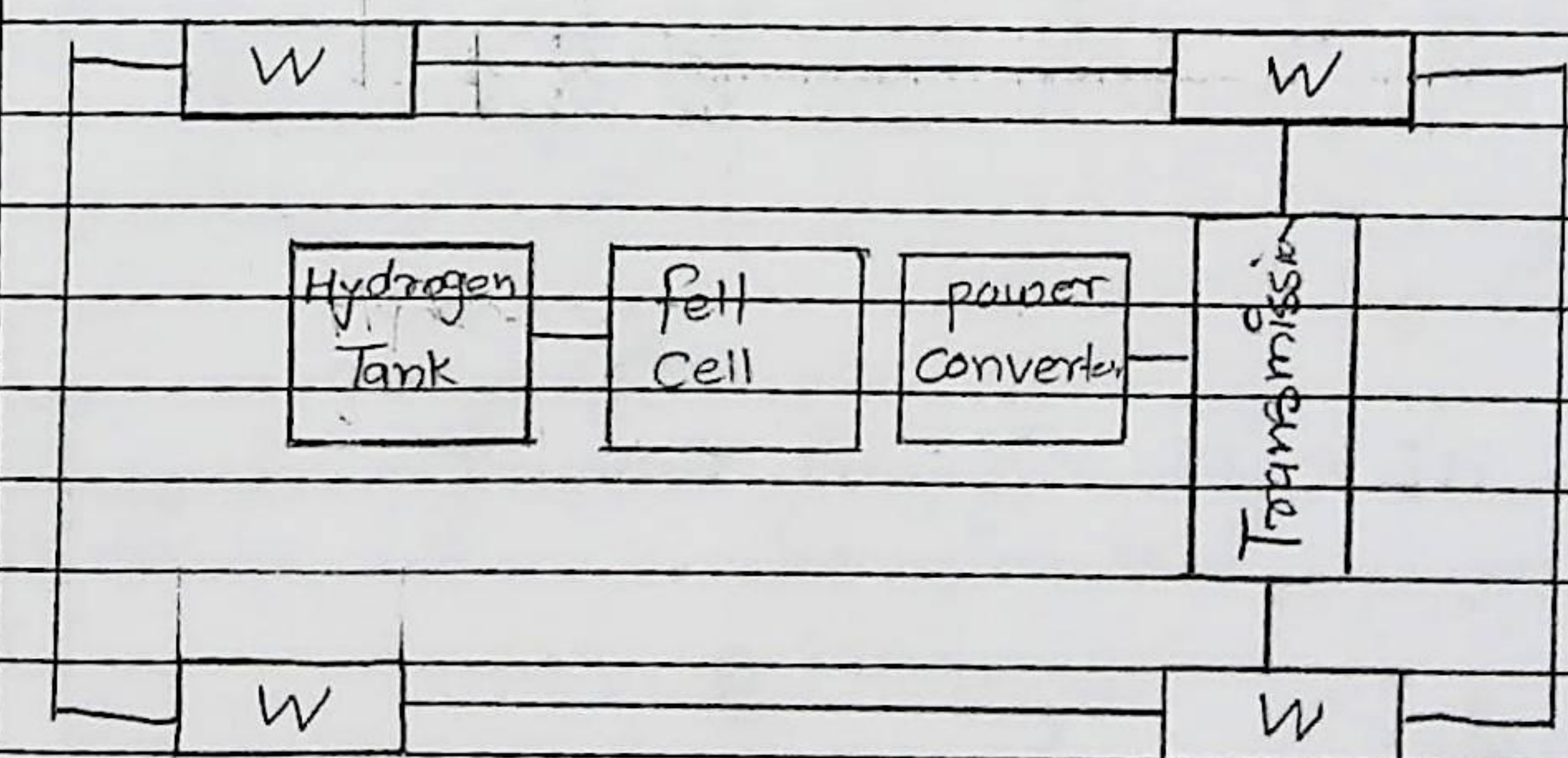
- electric motor as source energy pass the Transmission, Transmission power supply to wheels and vehicle is move.

- Batter Electric is low cost and Zero pollution.

b) Hybrid EVs



c) fuel cell electric vehicle



Q.No.

Examiner

Moderator

a b c d e =

Set. No. : P/Q/R/S

10

Section - II

Q. 5 a]

Answer

~~Answer~~

* Classification of HEVs

- a] Series (electric coupler)
- b] Paralla (Mech Coupler)
- c] Series - paralla
- d] Complex.

a] Series (electric coupler)

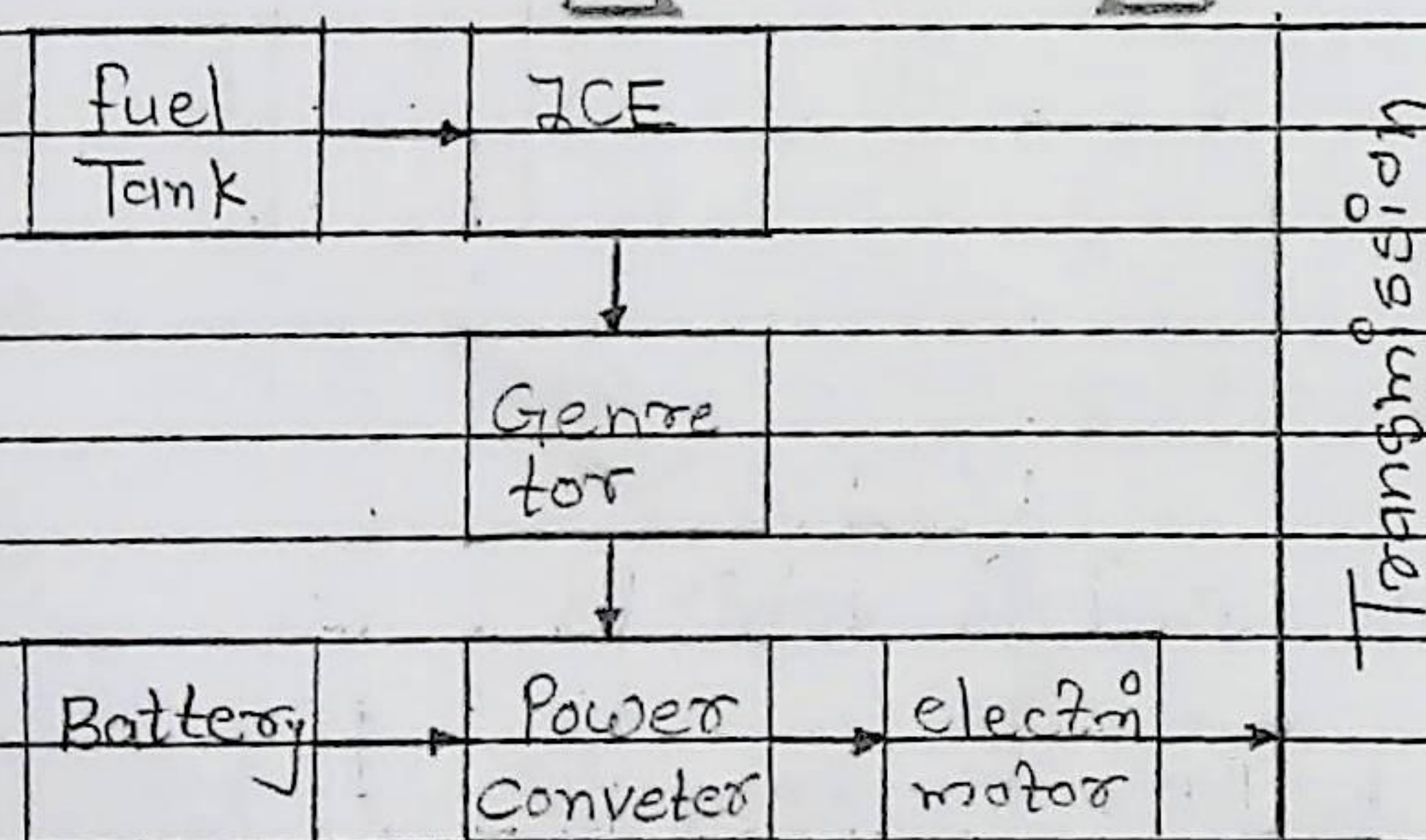


Fig No. 5a

— Series HEV as electric coupler.

Q.No.

Examiner

Moderator

a b c d e =

Set. No. : P/Q/R/S

11

— Comp:- Fuel Tank, ICE, Generator, Batter, Power converter, electric motor & Transmission

— fuel is pass on the engine and fuel is burned then power is generated in generator.

— Battery source is energy pass the power converter to AC to DC and DC to AC then electric motor then pass the power in Transmission and Transmission pass the wheels.

b] Paralla (Mech. Coupler)

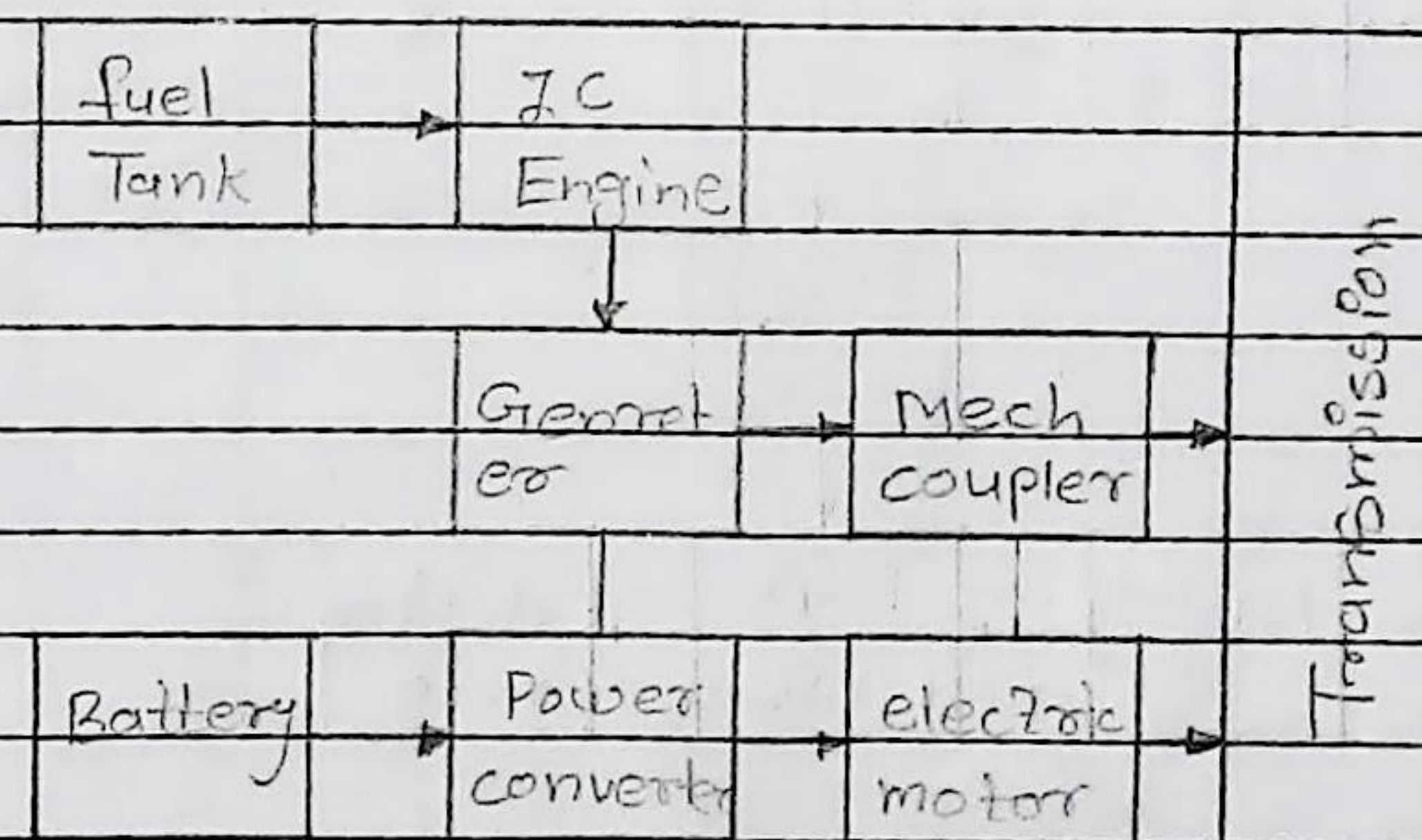


Fig No. 5b

— paralla HEV as mechanical coupler.



Q.No.

Examiner

a	b	c	d	e	=	

Moderator

Set. No. : P/Q/R/S

12

Components:- fuel Tank, ICE, Generator, Mech coupler, Battery, electric motor, Transmission

✓ (5.1) Fuel is inside the engine and engine outlet power is connect the Mech. coupler.

✓ Mech. coupler outlet is connect the Transmission and Power converter.

✓ Battery outlet is connect the power converter and power convert outlet elec. motor and electric motor outlet Transmission

c] Series - parallel

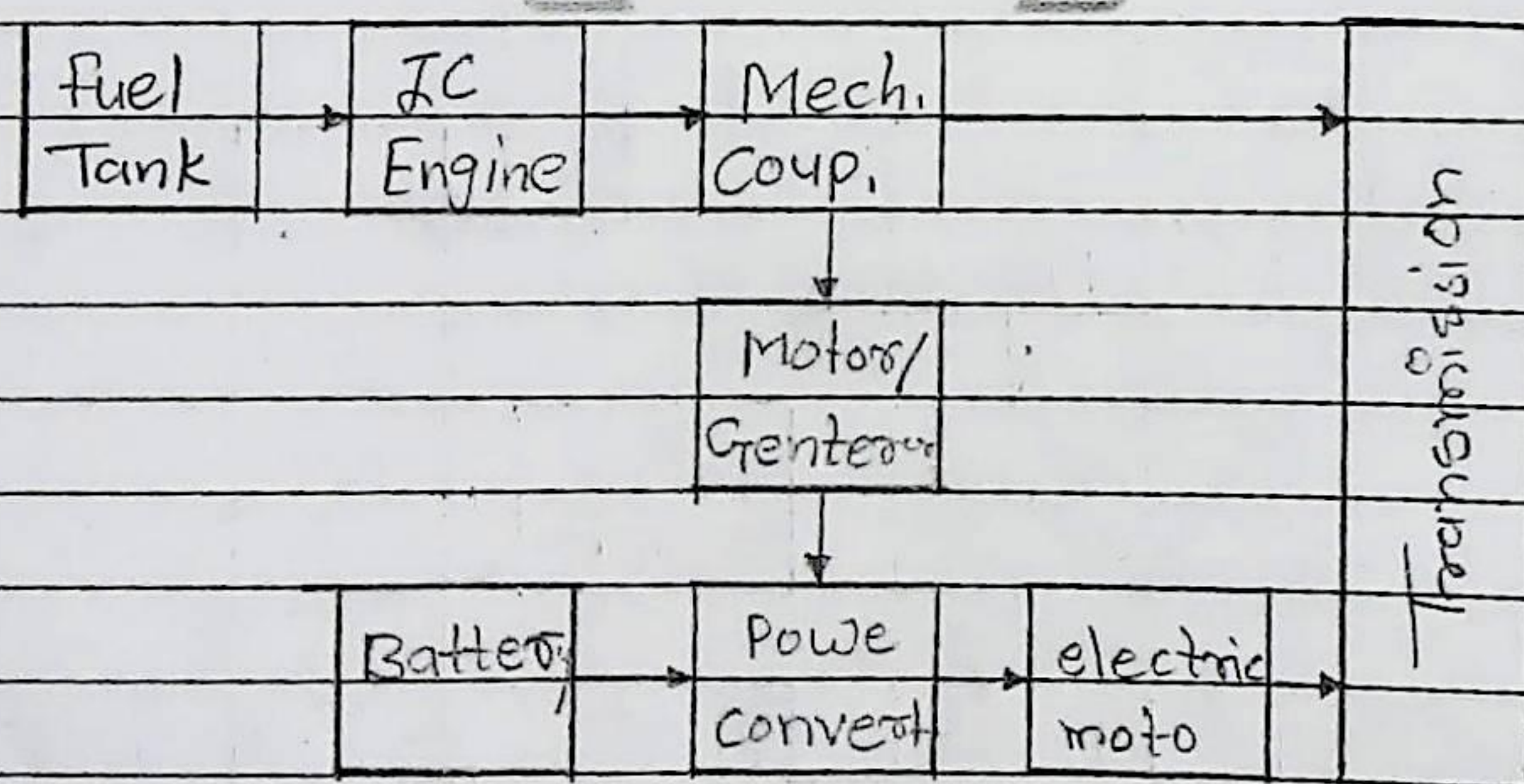


fig No. 5c

✓ (5.1) Components - fuel tank, IC Engine, Mech coupler, Motor/Generator, Battery power convert and Transmission.



Q.No.

Examiner

a	b	c	d	e	=	

Moderator

Set. No. : P/Q/R/S

13

✓ (5.1) Fuel is inside the engine & out is connect the mech coupler

✓ (5.1) Mech. coupler outlet is connect the Transmission and power converter

✓ Battery outlet is connect the power converter and power convert outlet elec. motor and electric motor outlet Transmission

d] Complex

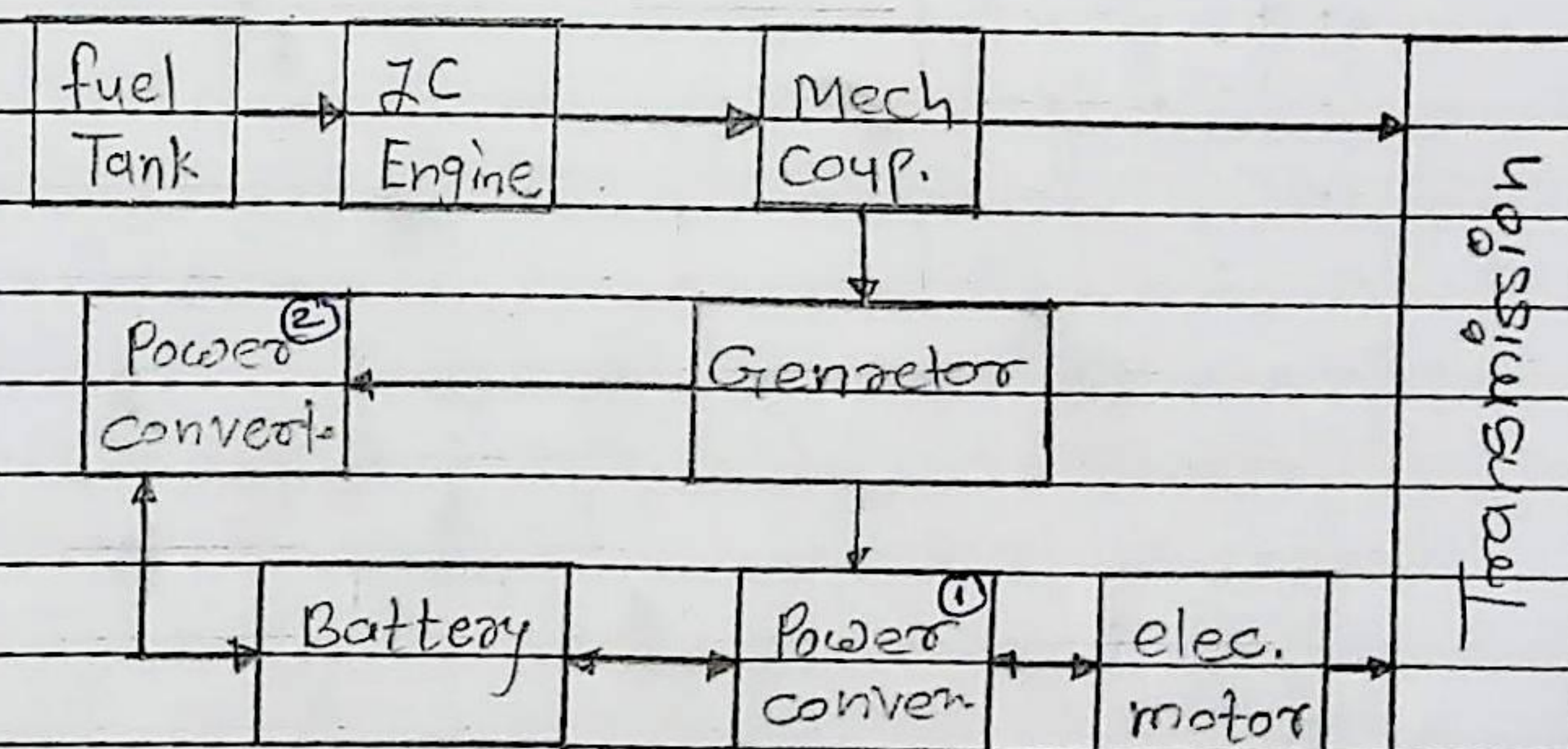


fig. No. 5d

Comp. - fuel Tank, IC Engine, Mech coupler, generator, power convert (1) power converter (2), Battery, electric motor.

Q.No.
Examiner
Moderator

a + b + c + d + e =
 Examiner
 Moderator
 Set. No. : P/Q/R/S

14

- Same prosiger as fig. 5a But this Type In 2 power conveter as use.
- power conveter load High then is used to 2 power conveter. for Re-generative Baking.

1/2 Q(4.2)

Q.No.
Examiner
Moderator

a + b + c + d + e =
 Examiner
 Moderator
 Set. No. : P/Q/R/S

15

Q. 4 e]

Q

Answer

or
No.

HEV

2 Q(4.5)

Fuel cell : EV

1. Energy Source is Batter and Fuel
2. Low cost per FCEV
3. HEV is charging
4. HEV is posible & more mobles.

Energy Source is Hydrogen.

High cost

FCEV is refilling

FCEV is not posible and less mobles

Q.No.
Examiner

Moderator

a	b	c	d	e	=	
					=	
					=	

Set. No. : P/Q/R/S

16

Q.4 b]

Answer

— a] ISE

b] Battery

c] Fuel Tank

d] Electric motor

e] Re-generative Braking system

f] Transmission

Q.No.
Examiner

Moderator

a	b	c	d	e	=	
					=	
					=	

Set. No. : P/Q/R/S

17

Q.4 d]

✓ Q(4.4)

a] Series (electric coupler)

b] parallel (Mech Coupler)

c] Series - parallel

d] complex

e] plug-in HEV

a] Series (electric coupler)

— as same is fig. No 5a (page No. 10)

b] parallel (Mech Coupler)

— as same is fig. No. 5b (page No 11)

c] Series - parallel

— as same is fig No. 5c (page No 12)

d] complex HEV

— as same is fig No. 5d (page No. 13)



Q.No.
[]

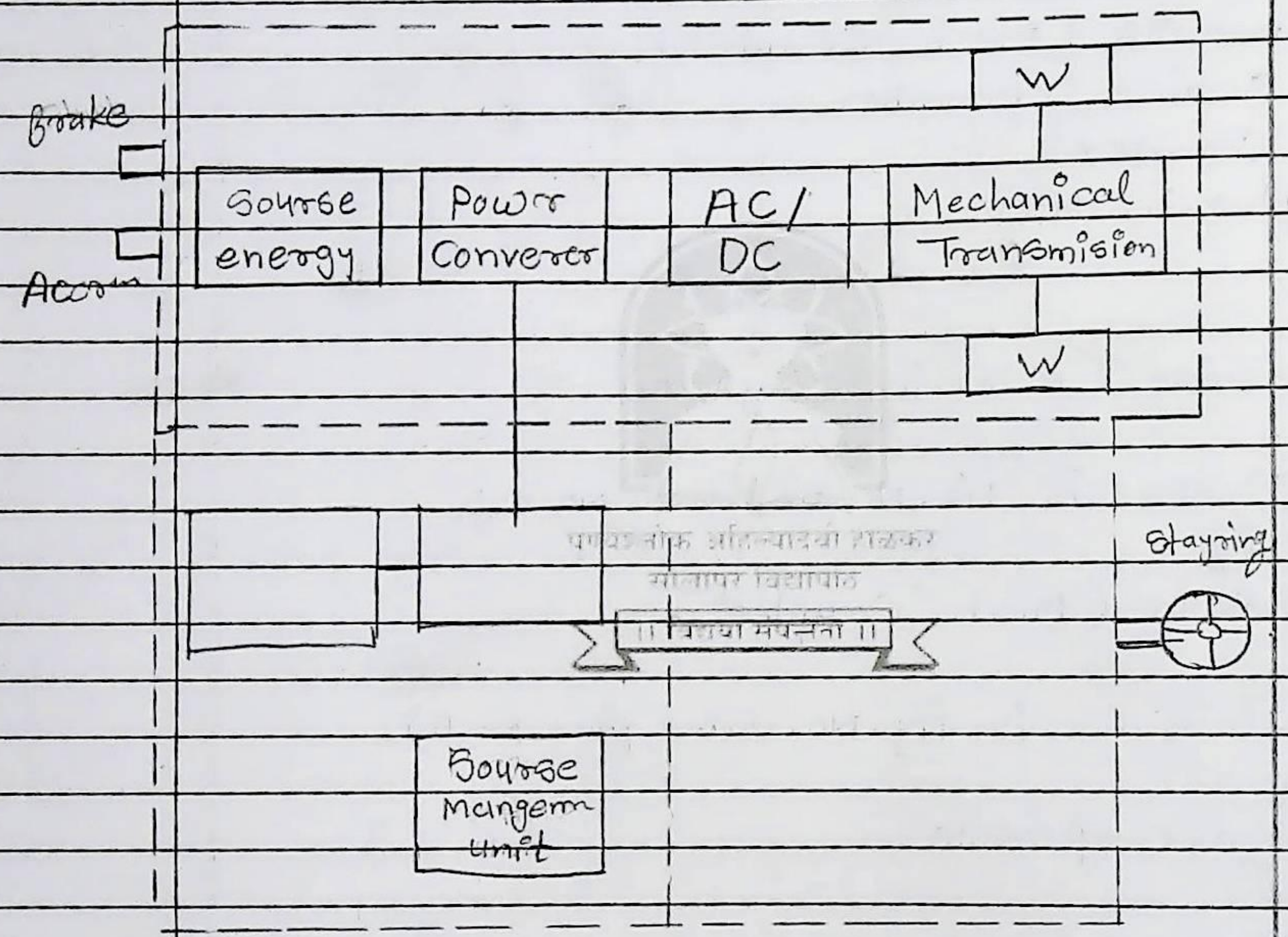
Examiner
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a [] + b [] + c [] + d [] + e [] = []
 Moderator [] + [] + [] + [] + [] = []
 Set. No. : P/Q/R/S []

18

Section - I

Q.3 C



Q(3.3)



Q.No.
[]

Examiner
[]

Q(3.3) a [] + b [] + c [] + d [] + e [] = []
 Moderator [] + [] + [] + [] + [] = []
 Set. No. : P/Q/R/S []

19

Q(3.3)



Q.No.

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Set. No. : P/Q/R/S

20



पुण्यन्याय अदिव्यादी राजा
मोलापर विद्यापीठ
॥ विद्या मयम् ॥

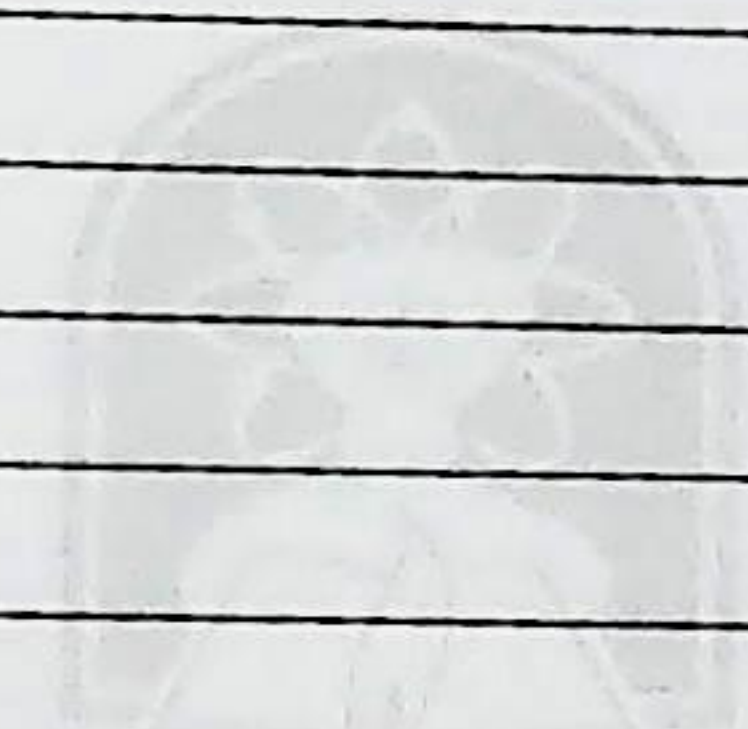
Q(3.3)



Q.No.

Examiner $\begin{matrix} a \\ \square \end{matrix} + \begin{matrix} b \\ \square \end{matrix} + \begin{matrix} c \\ \square \end{matrix} + \begin{matrix} d \\ \square \end{matrix} + \begin{matrix} e \\ \square \end{matrix} = \begin{matrix} \square \\ \square \end{matrix}$
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Set. No. : P/Q/R/S

21



पुण्यन्याय अदिव्यादी राजा
मोलापर विद्यापीठ
॥ विद्या मयम् ॥

Q(3.3)



Q.No.

Examiner

a	b	c	d	e	
+	+	+	+	+	

Moderator

Set. No. : P/Q/R/S

22

Q(3.3)



Q.No.

Examiner

a	b	c	d	e	
+	+	+	+	+	

Moderator

Set. No. : P/Q/R/S

23



Q.No.

Examiner

a	b	c	d	e	

Moderator

Set. No. : P/Q/R/S

24

Q(3.3)

पञ्चमहाक अतिन्यायका होलकर

मानापुर विद्यालय

॥ विद्यया मय-नता ॥